Weight-normalized Pharmacokinetics with Context-sensitive Metrics

Adults (male, 40 y, 70 kg, 170 cm); model-derived values for comparative illustration.

Panel A. Pharmacokinetic Parameters and Effect-site Equilibration (k_{e0})

Drug	Cl (mL/kg/min)	V1 (L/kg)	V2 (L/kg)	V3 (L/kg)	k_{10} (min^{-1})	k_{12} (min^{-1})	k_{21} (min^{-1})	k_{13} (min^{-1})	k (min^{-}
Propofol	30	0.2	1.8	2	0.15	2.5	0.28	2	0
Dexmedetomidine	9	0.2	1	1.5	0.04	1	0.2	1	0.1
Fentanyl	13	0.25	1.7	2	0.05	0.48	0.07	0.4	0.0
Hydromorphone	8	0.3	1.5	2	0.03	0.33	0.07	0.27	0.0
Remifentanil	40	0.1	0.5	0.8	0.4	1.5	0.3	1.2	0.1
Alfentanil	6	0.3	1.2	1.5	0.02	0.33	0.08	0.27	0.0
Methadone	1.5	0.4	2	3	0	0.12	0.03	0.1	0.0
Rocuronium	5	0.2	1	1.5	0.02	0.5	0.1	0.4	0.0
Vecuronium	4	0.2	1	1.5	0.02	0.4	0.08	0.3	0.0
Succinylcholine	20	0.2	0.8	1	0.1	1	0.25	0.75	0.1
Midazolam	9	0.15	1.2	1.5	0.06	1.07	0.13	0.8	0.0
Diazepam	0.5	0.5	2.5	3	0	0.1	0.02	0.08	0.0

Panel B. Half-lives, Plasma CSHT-50 (by context), and Effect-site Context-sensitive Decrement (1h)

Drug	$t_{1/2}$ elim (min)	$t_{1/2}$ fast dist (min)	$t_{1/2}$ slow dist (min)	CSHT-50 (10m) (min)	CSHT-50 (1h) (min)	CSHT-50 (4h) (min)
Propofol	4.62	0.28	2.49	45	83.7	90.2
Dexmedetomidine	15.4	0.69	3.46	64.2	179	201
Fentanyl	13.3	1.44	9.82	1.62	134	191
Hydromorphone	26	2.08	10.4	2.2	193	293
Remifentanil	1.73	0.46	2.31	7.92	20.7	22.2
Alfentanil	34.6	2.08	8.32	2.9	238	318
Methadone	185	5.54	27.7	3.84	140	1.72e+03
Rocuronium	27.7	1.39	6.93	2.27	258	344
Vecuronium	34.6	1.73	8.66	2.18	281	418
Succinylcholine	6.93	0.69	2.77	27.1	61.3	67.1
Midazolam	11.6	0.65	5.2	7.19	171	207
Diazepam	693	6.93	34.6	4.72	233	5.7e+03

Models/Sources. Propofol: Schnider et al. (Anesthesiology 1998/1999); Remifentanil: Minto et al. (Anesthesiology 1997); Fentanyl: Shafer et al. (Anesthesiology 1990/1991); Alfentanil: Scott & Stanski (Anesthesiology 1987–1990); Dexmedetomidine: Ebert/Talke 2000; Iirola 2011 (Anesthesiology); Hydromorphone: Rose 1996 (Br J Anaesth); Lötsch 2002 (Clin Pharmacokinet); Methadone: Ferrari 2004 (Anesthesiology); Eap 2002 (Clin Pharmacokinet); NMBAs: Miller's Anesthesia; Magorian/Bevan 1993 (Anesthesiology).

Notes. Numeric values here are illustrative until parameters are directly extracted from the cited models for the specified covariates. All values are weight-normalized (Cl, Q in mL/kg/min; V in L/kg). CSHT and effect-site metrics simulated with a 1 unit/kg/min infusion and abrupt stop; effect-site via $\frac{dC_e}{dt} = k_{e0}(C_1 - C_e)$. Use model-specific coefficients for clinical work.